

MUST lecture 12

Grouping / clustering of multivariate observations based on distance metrics

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Grouping / clustering of multivariate observations based on distance metrics

Problem Given observations $X_1, \dots, X_n$ $X_j \in \mathbb{R}^p$
Split observations in K disjoint clusters based on similarity/distance between obs
(K must usually be assumed or iterated)

Purpose Interpretation / understanding of data
Potentially preanalysis for LDA/QDA

Standardization of data if units or dynamic range very different for variables (zscore)

Distance between observations m-norm $d(x, y) = \left(\sum_{i=1}^p |x_i - y_i|^m \right)^{\frac{1}{m}}$

Distance between clusters Linkage metric (see MATLAB demo)

Algorithms Hierarchical clustering
K-means clustering
Gaussian Mixture Model clustering
(details in pdf-file "Clustering Algorithms" and on blackboard)